

nnFoundation

3D foundation models for radiology



Original paper arxiv.org/abs/2609.26924

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Default, from scratch, pretrained

	Network	Starting weights
■ nnU-Net Default	standard nnU-Net U-Net	random
■ From scratch	ResEnc-L or Primus	random
■ Pretrained	ResEnc-L or Primus	MAE-pretrained encoder (2.1M unlabeled scans)
All three	then trained on the same labeled data	

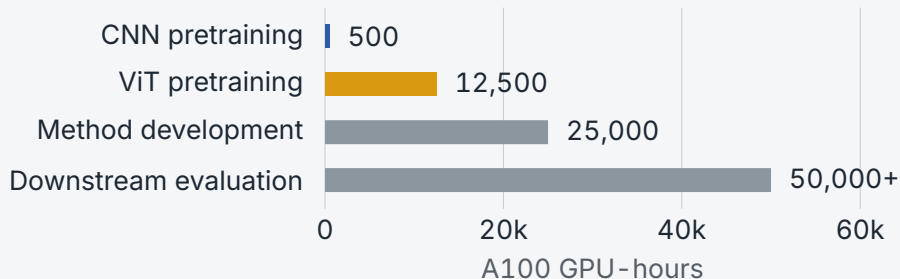
Two models, one recipe

Model	Backbone	Params	Pretraining	A100 hours
■ nnF-CNN	ResEnc-L	102M	MAE, 75% masked	500
■ nnF-ViT	Primus	674M	MAE, 80% masked	12,500

Shared setup

Pretraining loss	MSE, masked voxels
Input patch	192 ³
Training steps	625,000
Batch size	8 (CNN), 96 (ViT)

Compute



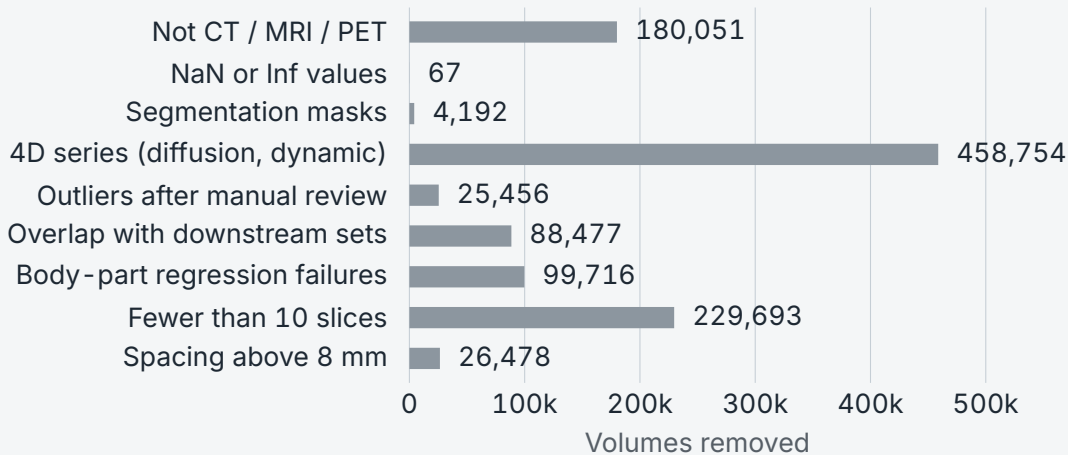
GPU	NVIDIA A100	Clusters	HoreKa, JUWELS
ViT pretraining run	96 GPUs × 130 h	Fine-tuning GPU	1 A100 40GB
Budget per SSL method	400 h	Fine-tuning batch	2

Pretraining data vs evaluation

Body region	Pretraining	Evaluation
Brain	290K	29.3K
Head & neck	310K	2.5K
Chest	390K	78.8K
Abdomen	550K	35.8K
Whole body / MSK	370K	13.1K
Unknown	260K	–
Total volumes	2.16M	160K

Pretraining source	Volumes
Clinical partners	1,315,227
Research cohorts	737,714
Small public sets	105,629
MRI	71.7%
CT	26.6%
PET	1.7%

From 3.27M to 2.16M volumes



Pretraining scale vs prior models

Model	Input	Modality	3D volumes (equivalent)	Eval datasets
■ nnFoundation	3D	CT, MRI, PET	2,100,000	108
CURIA	2D	CT, MRI	1,150,000 [†]	23
Prima	3D	MRI	221,147	10
VoCo	3D	CT	160,000	48
CT-FM	3D	CT	148,000	9
MIS-FM	3D	CT	110,000	3
Merlin	3D	CT	25,528	8
VISTA3D	3D	CT	11,454	19
SwinUNETR	3D	CT, MRI	5,050	11

[†] 150K volumes + 200M 2D slices counted as slices $\div$ 200

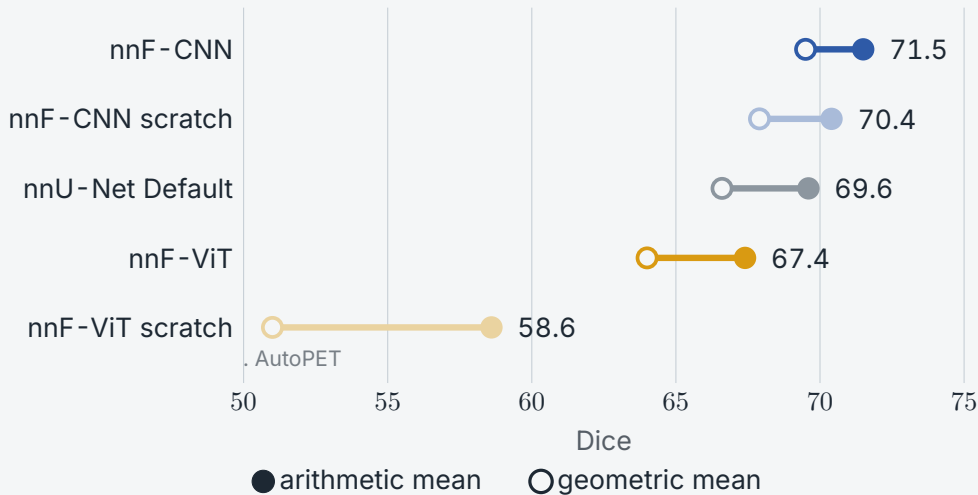
Picking the pretraining objective

630K public volumes, 9 dev tasks	Seg Dice (5)	Cls AUROC (2)	Det mAP (2)
■ CNN + MAE	77.5	64.8	37.9
■ CNN + contrastive	76.0	70.8	34.6
■ ViT + MAE	75.6	73.2	36.7
■ ViT + DINOv2	73.1	70.1	22.2
■ ViT + contrastive	71.7	69.3	30.4

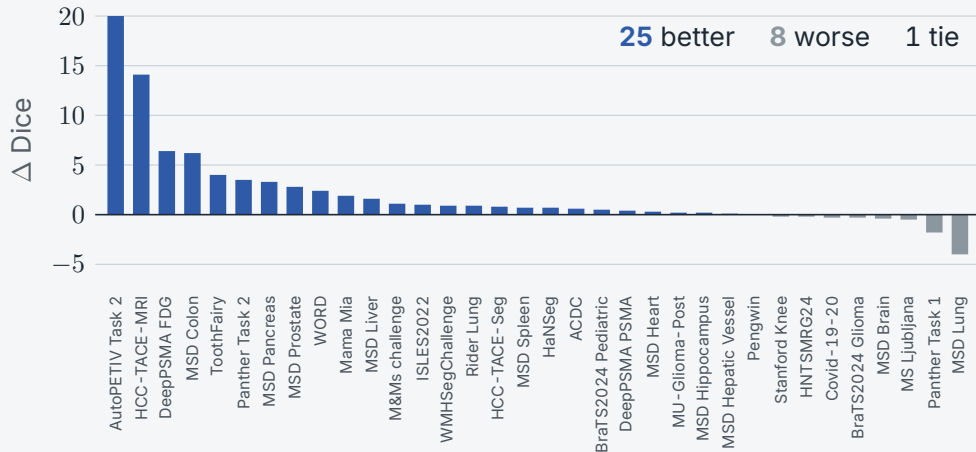
630K vs 2.1M volumes

Same 9 dev tasks	Data	Seg Dice	Cls AUROC	Det mAP
■ CNN + MAE	630K	77.5	64.8	37.9
■ nnF-CNN small (released)	2.1M	77.5	64.1	36.2
■ nnF-CNN big	2.1M	76.7	67.7	39.2
■ ViT + MAE (Primus-M)	630K	75.6	73.2	36.7
■ nnF-ViT small	2.1M	73.7	69.2	34.1
■ nnF-ViT big (released)	2.1M	77.1	74.1	36.7

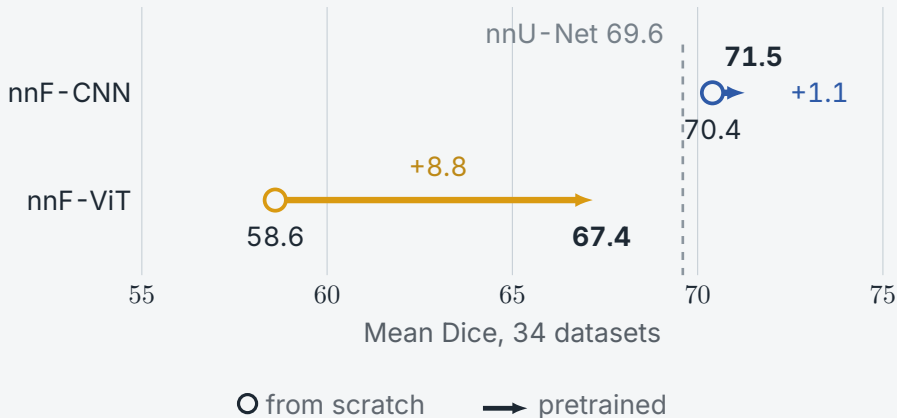
Segmentation, 34 datasets



nnF-CNN vs nnU-Net, per dataset



Pretrained vs from scratch



Topology adaptation

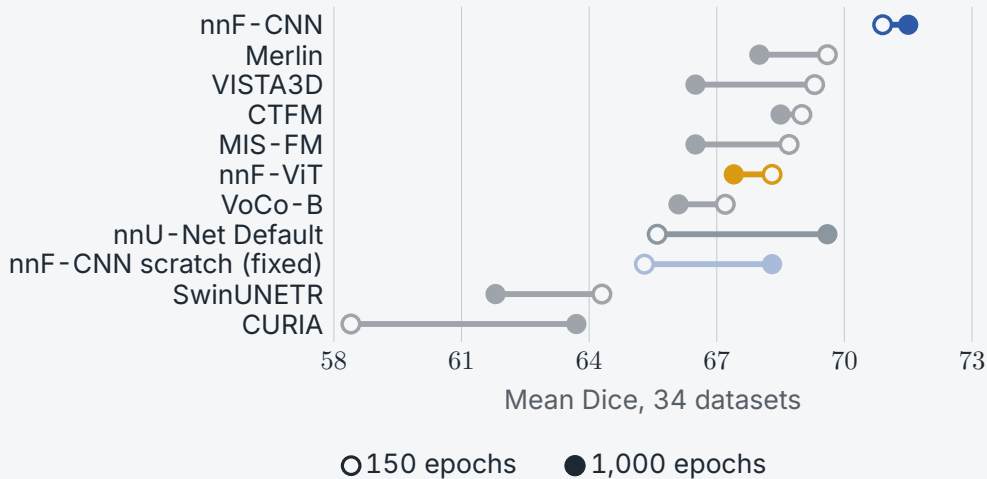
Seg Dice, 34 sets	Fixed	Planned
■ From scratch	68.3	70.4
■ Pretrained	70.3	71.5

Det mAP, 9 sets	Fixed	Planned
■ From scratch	68.6	69.6
■ Pretrained	71.4	72.6

Six datasets, every model

Dice	MSD Pancreas	MSD Lung	ACDC	BraTS Glioma	WORD	AutoPET IV
 nnF-CNN	67.0	66.2	91.7	73.2	85.5	41.2
 nnF-CNN scratch	63.7	68.0	91.2	73.2	85.7	37.5
 nnU-Net Default	63.7	70.2	91.1	73.5	83.1	21.2
 nnF-ViT	53.9	61.6	88.2	71.5	85.4	27.0
 nnF-ViT scratch	20.6	61.6	87.4	67.3	42.0	0.0
CTFM	56.8	68.1	91.1	73.7	85.6	33.2
VISTA3D	58.8	36.8	90.0	74.3	84.5	38.8
MIS-FM	60.2	37.5	90.9	72.8	85.5	41.2
Merlin	56.3	67.4	90.4	73.8	85.2	32.8
VoCo-B	57.9	61.0	89.2	73.2	84.3	31.7
CURIA	53.3	48.4	88.4	58.7	81.6	23.0
SwinUNETR	26.3	36.6	87.9	72.1	81.6	22.6

150 vs 1,000 fine-tuning epochs



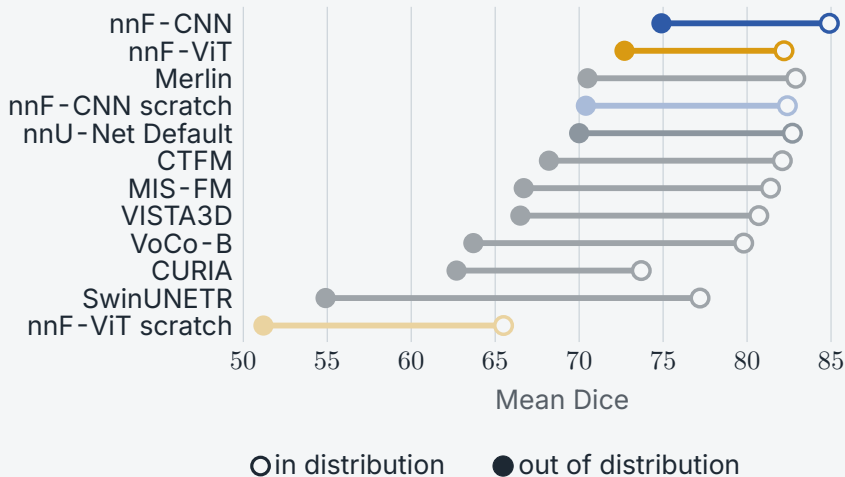
Low-data segmentation



Low-data: the six datasets

Dataset	Scan and target	Dev set	Δ Dice vs scratch			
			10 cases	20	40	100
Mama Mia	MRI, breast tumour	no	+6.1	+4.3	+4.5	+1.7
CTSpine1K	CT, vertebrae	no	+2.4	+6.8	+3.7	+7.3
HNTSMRG24	MRI, head & neck tumour	no	-2.2	-1.8	-0.5	-0.3
KiTS	CT, kidney tumour	yes	+6.7	+4.0	+2.5	+1.0
AMOS	CT + MRI, abdomen organs	yes	+2.1	+1.0	+0.7	+0.3
BraTS 2024 Glioma	MRI, brain tumour	no	+2.8	+2.9	+2.8	+2.7
■ Mean			+3.0	+2.9	+2.3	+2.1

Out of distribution



External partners

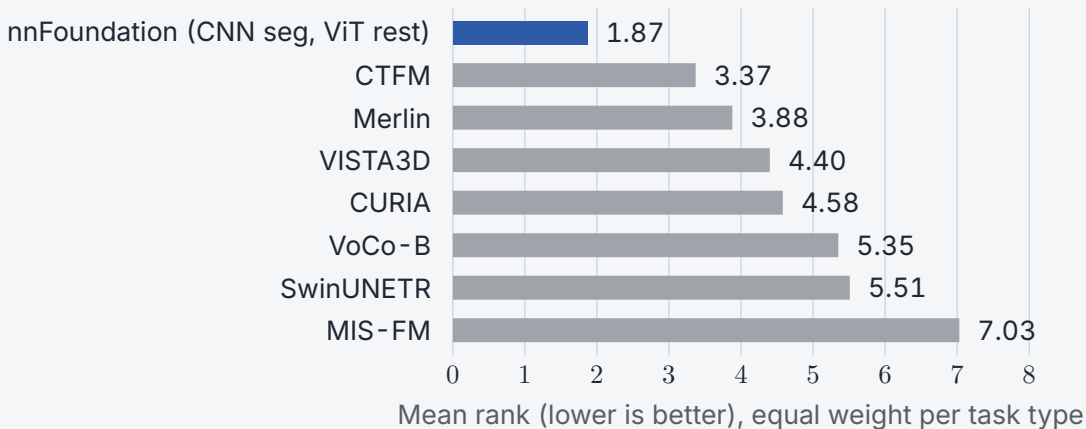
Segmentation, Dice	Datasets	Scratch	■ nnF-CNN	Δ	CI above 0
10% of training data	25	59.5	63.7	+4.2	yes
20%	26	70.7	72.5	+1.9	yes
50%	28	74.9	75.8	+0.9	no
100%	28	76.5	77.7	+1.1	yes

Lung nodule detection (NLST), mAP	Scratch	■ nnF-CNN	Δ	CI above 0
1% of training data	40.7	43.9	+3.2	no
10%	54.9	56.1	+1.1	no
100%	64.8	63.8	-1.0	no

Best model by task

Task	Datasets	Best	Margin
Segmentation	34	■ nnF-CNN	+1.9 Dice over nnU-Net, significant
Detection	9	■ nnF-CNN	no external baselines
Classification	8	■ nnF-ViT	tied with CTFM, CURIA, VISTA3D
Retrieval	5	■ nnF-ViT	first on 1 of 5 (tumour stage AP)
Report generation	2	■ nnF-ViT	F1 43.7 vs 42.8 (CURIA)
Out of distribution	7	■ nnF-CNN	74.9 vs 70.4 Dice from scratch

Overall rank across task types



Beyond radiology: microscopy

Dice	Val. cases	CNN scratch	nnF-CNN	ViT scratch	nnF-ViT
Airway (light sheet)	4	86.4	88.7	75.5	90.4
CREMI synapses (EM)	3	40.6	41.5	35.2	36.1
Urocell mitochondria (EM)	1	72.4	73.5	66.3	73.7
SELMA3D (light sheet)	9	71.5	72.3	70.3	71.3
LAPD mouse (fluorescence)	7	96.5	96.9	96.7	97.2

What is in the paper

	Included
Semantic segmentation	yes
Detection, boxes (Deformable DETR)	yes
Instance segmentation	no
MedNeXt baseline	no
STU-Net baseline	no
nnU-Net ResEnc-L baseline	no
Scale / modality ablations	no
nnU-Net baseline in low-data experiments	no

No instance-level evaluation

Searched in paper and supplement	Mentions	Where
Panoptic quality (PQ)	0	–
Instance segmentation	0	only in a cited title (RibFrac, ref. 84)
Lesion-wise / per-lesion metrics	0	–
Object-level F1	0	F1 appears only for report labels
Connected components	1	retrieval crop preprocessing
Segmentation metrics used		Dice, NSD (voxel-level, per class)
Detection metrics used		mAP, FROC, AP@IoU 0.10 (boxes)

Where text and tables disagree

	Main text	Supplement or code
nnF-CNN in-distribution Dice	84.4	84.9 (Table F32)
nnF-CNN OOD retention	88.8%	88.2%
Highest retention	nnF-CNN	nnF-ViT, 88.5%
CURIA in-dist. Dice / retention	71.7 / 87.5%	73.7 / 85.1%
Merlin in-dist. Dice / retention	81.4 / 86.7%	82.9 / 85.1%
"NoMask" loss	masked voxels only	whole volume (nnssl code)
CNN scratch at 1,000 epochs	70.4 (Fig. 2)	68.3, fixed topology (F26)

Verdict

Holds up

108 tasks, bootstrap CIs
Shows nnU-Net beats every prior FM
Reports its own negative transfer
Drops into nnU-Net today
Dynamic topology: +1.2 Dice, +1.2 mAP
Big gains with few labels or short training

Best out-of-distribution Dice

Weaker than it sounds

Pretraining adds +1.1 Dice for the CNN
Baseline is nnU-Net Default, not ResEnc
Some baselines broke in their pipeline
Classification and retrieval wins are soft
No ablation of what drives the gain
CNN gained nothing from 630K to 2.1M on dev tasks

Several baselines score higher at 150 than 1,000 epochs

Code and weights

What	Where
Pretraining code	github.com/MIC-DKFZ/nnssl
■ CNN weights	huggingface.co/MIC-DKFZ/nnFoundationCNN
■ ViT weights	huggingface.co/MIC-DKFZ/nnFoundationViT
Segmentation fine-tuning	nnU-Net, finetuning_from_nnssl_checkpoints.md
Detection, classification, reports	not released yet
2.1M pretraining corpus	not released (mostly private)